

AKNUR SHERKHAN

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EDUCATION

Minerva University

San Francisco, CA

Candidate for Bachelor's in Computational and Social Sciences (Data Science and Economics)

Expected Graduation May 2027

- Relevant coursework: Machine Learning, Simulation & Modeling, Data Structures & Algorithms, Bayesian Statistics, Probability & Statistics, Econometrics, Linear Algebra, Multivariable Calculus.

EXPERIENCE

Puerta 18

Buenos Aires, Argentina

CRM and Data Systems Developer

Sep 2025 - Apr 2026

- Built the organization's first data infrastructure from scratch: digitized 18 years of unstructured paper records into a centralized SQL system; defined data standards and documentation for long-term maintainability.
- Engineered automated reporting pipelines (Python, SQL) and live KPI dashboards used daily by staff; eliminated 15+ hrs/week of manual reporting and increased admin capacity by 0.4 FTE.

Terra Education at UC Berkeley

Berkeley, CA

Data Science and ML Teaching Assistant

Jun 2025 - Aug 2025

- Taught an ML curriculum to 150+ high school students covering regression, classification, and bias-variance tradeoffs.
- Supervised end-to-end student projects: data cleaning, feature engineering, model training, evaluation, and reporting.

GSP Cloud

Seoul, South Korea

Product Development Intern - ML and Applied AI

Jan 2025 - May 2025

- Built a GPT-4-powered agentic learning assistant (React.js + Python) for a Korean B2B edtech platform; implemented prompt chaining, tool-augmented API calls, and multi-step reasoning to summarize lectures, generate timestamp-referenced answers, and create real-time comprehension quizzes.
- Designed and executed A/B tests across ~400 users on AI response strategies; analyzed behavioral logs to identify engagement drop-off patterns, driving a 12% increase in user retention.
- Evaluated GPT-4o vs. GPT-4 across 50+ test prompts; adapted system prompts to preserve stable behavior under model updates, documenting regression-test protocol for ongoing model migrations.

Minerva University

San Francisco, CA

Data Analysis Intern - Admissions Team

Sep 2024 - Present

- Analyzed 20,000+ applicant records using Python (Pandas, NumPy) to identify trends in applicant quality and volume; built segmentation logic that directly guided team prioritization during peak review cycles.
- Built a Gemini API-powered web app to surface rubric-aligned observations on applicant essays, enabling reviewers to process 30% more applications per session without sacrificing evaluation quality.
- Developed Python-based QA scripts to resolve data inconsistencies across 3000+ applications, maintaining high data integrity.

PROJECTS

[Kazakh-Russian Speech Classifier](#) | *PyTorch, BiLSTM, Bahdanau Attention, librosa, scikit-learn*

- 89% test accuracy on a low-resource, bilingual audio classification task: built an end-to-end pipeline on 4,800+ audio chunks using an Attentional BiLSTM with Bahdanau attention.
- Resolved overfitting by integrating Google FLEURS corpus (300+ speakers), improving generalization by ~27%.
- Implemented a Seq2Seq autoencoder with teacher-forcing decay for unsupervised representation learning.

[Personalized Photo Recommendation Model](#) | *OpenAI CLIP, KMeans, Cosine similarity, Revealed Preference Theory*

- Modeled individual Instagram aesthetic using CLIP embeddings + cosine similarity to rank camera-roll photos by predicted posting likelihood; framed as a revealed preference problem where past posts encode latent taste.
- Clustered posting history via KMeans to identify sub-aesthetic profiles; system generalizes to new photos without retraining, allowing real-time ranking.

[Urban Traffic Congestion Simulation](#) | *Python, NumPy, OpenStreetMap, Monte Carlo, Network Analysis*

- Built an OOP traffic simulator using real OpenStreetMap road networks; modeled congestion via the Greenshields speed-density model with dynamic rerouting across 50-run Monte Carlo evaluations.
- Identified edge betweenness centrality as a dominant structural predictor of congestion (Spearman $\rho = 0.93$), validated across Berlin and Buenos Aires road networks.

SKILLS

- **Languages:** Python (advanced), SQL (advanced), JavaScript/React (intermediate), R (intermediate)
- **ML & Modeling:** PyTorch, TensorFlow, scikit-learn, CLIP/ResNet, LLM integration, Prompt engineering, Seq2Seq
- **Data & Analysis:** Pandas, NumPy, Matplotlib, Tableau, A/B testing, Experiment design, Causal inference, MC simulation